Assessment-Bound Disparities: Divergence Between National Evaluation Submissions and Index-Based Bibliometrics in Australia's ERA 2018

Lawrence Cram[1*]

[1*]College of Science, The Australian National University, Canberra, 2601, ACT, Australia.

Corresponding author(s). E-mail(s): lawrence.cram@anu.edu.au;

Abstract

National research evaluation exercises routinely compile massive, curated corpora of scholarly outputs to assess institutional performance. In Australia's Excellence in Research for Australia (ERA) 2018, the 40 eligible Higher Education Providers (HEPs) submitted 567,647 outputs, reconciling into 455,741 distinct canonical works published between 2011 and 2016. A pervasive assumption in evaluative bibliometrics is that an assessment-defined corpus mirrors the scholarly record indexed under institutional affiliation addresses. Linking ERA submissions against OpenAlex, we find deep, structural divergences between the two corpora. Across 2011–2016, OpenAlex records 744,574 works bearing an Australian HEP author affiliation, of which only 401,261 (53.9%) appear in ERA; 343,313 works (46.1%) were not submitted. While 75,348 unsubmitted works represent document types formally excluded by assessment rules (preprints, meeting abstracts, book reviews, editorials), over 222,000 are standard peer-reviewed journal articles, books, and book chapters. Crucially, 45,822 of these unsubmitted Australian works are directly cited by ERA papers (generating 109,443 citation links), including landmark methodological contributions recording thousands of citations. Furthermore, among submitted works, the institution claiming the paper in ERA matches the published institutional address on the work in only 67.6% of cases, with 13.8% (49,962 works) bearing zero Australian institutional affiliation. This institutional attribution divergence exhibits a strict monotonic temporal gradient across the six-year retrospective window, with purely foreign affiliations rising from 8.4% in 2016 to 19.4% in 2011. Grounded in Olensky's framework of bibliometric data accuracy, we demonstrate that these disparities arise from the structural divergence between two fundamentally different bibliometric constructs: evaluating the retrospective output of an institutional workforce fixed at

a single census date versus measuring the scholarly output produced within an institutional address.

Keywords: Scientometrics, Bibliometrics, Research evaluation, Excellence in Research for Australia, Data accuracy, Citation matching, Institutional affiliation

1 Introduction

Performance-based research funding systems evaluate institutional research capacity through statutory submission exercises, benchmarking research volume, citation density, and disciplinary concentration [**?**]. In evaluative bibliometrics and university ranking frameworks, an operational assumption posits that an assessment-defined corpus aligns with the publication portfolio retrieved when querying global bibliographic databases by institutional affiliation addresses and publication year [**? ?**]. Discrepancies between submission inventories and index-based corpora are routinely categorized as coverage deficits, database inaccuracies, or transcription noise [**? ?**].

Olensky established that bibliometric data quality is governed by three distinct structural tiers: input data accuracy at source, algorithmic entity resolution and citation matching, and corpus definition boundaries [**?**]. When evaluating national research corpora, corpus definition boundaries diverge along administrative lines. National research evaluation frameworks—exemplified by the UK Research Excellence Framework (REF) and Australia's Excellence in Research for Australia (ERA)—define publication eligibility by staff employment at an administrative census date rather than the institutional affiliation printed on the publication at the time of authorship [**?**].

In Australia's ERA 2018 assessment, 40 Higher Education Providers (HEPs) submitted 567,647 research outputs across four traditional categories (journal articles, conference papers, books, and book chapters) and non-traditional research outputs published between 2011 and 2016. Deduplication and transitive identifier clustering reconcile these submissions into 455,741 distinct canonical works. Querying the OpenAlex global bibliographic database [**?**] across the identical 2011–2016 window for authors affiliated with these 40 institutions retrieves 744,574 works, indicating a gross numerical divergence of 288,833 works.

This paper establishes the mathematical volume, document type composition, citation integration, and institutional attribution divergence between the assessment-defined ERA 2018 corpus and the address-defined OpenAlex corpus. We resolve three formal empirical questions:

1. What is the exact volume, document type distribution, and citation density of publications carrying Australian university affiliations in OpenAlex (2011–2016) that are absent from ERA submissions?
2. What proportion of these omitted works are directly cited by the submitted ERA corpus, and how central are they to the internal citation graph?

3. What proportion of submitted works diverge from the submitting university in their printed affiliations, and how does this attribution divergence scale across the six-year retrospective window?

2 Methodology

2.1 The ERA 2018 Assessment Corpus

The Australian Research Council (ARC) mandated that eligible Higher Education Providers (HEPs) submit all research outputs authored by eligible academic staff employed at the census date of 31 March 2017 and published within the six-year retrospective window from 1 January 2011 to 31 December 2016. Across 40 participating HEPs, the raw submission inventory contains 567,647 output records spanning four traditional categories (journal articles, conference papers, books, book chapters) alongside non-traditional research outputs (NTROs). Output records record titles, publication years, output categories, publication outlets, digital object identifiers (DOIs), and institutional identifiers, omitting author lists.

Deduplication across institutions utilizes a transitive connected-component graph constructed via Disjoint Set Union (DSU). Equivalence edges connect records sharing normalized DOIs, publisher proprietary IDs, or exact normalized titles matched within publication years. This procedure collapses 567,647 institutional submissions into 455,741 canonical works, identifying 83,945 collaborative outputs submitted independently by two or more Australian universities.

2.2 Multi-Stage Linkage to OpenAlex

Canonical ERA outputs are linked to the global OpenAlex snapshot (October 2026 release, containing 1.9×10^8 works and 1.4×10^7 authorships) using a deterministic three-stage matching protocol:

1. **Compact Direct Match**: Exact string normalization of DOIs against primary OpenAlex work entities resolves 530,978 submissions.
2. **Alternate Location and XPAC Match**: Querying secondary identifiers and alternate repository records stored across the 1,898 partitions of OpenAlex's `work_locations` table recovers 39,152 additional submissions.
3. **Canonical Synthesis**: Merging output-level matches to canonical work clusters produces 427,882 matched canonical works, establishing a 93.9% overall linkage rate and a 98.5% linkage rate for journal articles.

2.3 Extraction of the Unsubmitted Australian Corpus

The unsubmitted reference corpus is extracted from OpenAlex's core tables (`compact/works` and `compact/authorships`). Extraction requires satisfaction of three formal constraints:

$$\text{Publication Year: } 2011 \leq \text{py} \leq 2016 \tag{1}$$

$$\text{Institutional Affiliation: } \exists a \in \text{Authorships}(w) : \text{Affiliation}(a) \in \mathcal{H}_{\text{AU}} \tag{2}$$

$$\text{Assessment Exclusion: work_idx}(w) \notin \mathcal{W}_{\text{ERA}} \tag{3}$$

where $\mathcal{H}_{\text{AU}}$ denotes the set of 40 Australian universities registered in OpenAlex's institution registry (country code `AU`, entity type `education`), and $\mathcal{W}_{\text{ERA}}$ denotes the set of OpenAlex works matched to the canonical ERA dataset.

2.4 Internal Citation Graph Extraction and Affiliation Concordance

Citation links originating from ERA-submitted works into unsubmitted works are extracted from OpenAlex's directed citation network (`compact/references`, 12.15 GB). An internal citation link is defined by a directed pair (u, v) where $u \in \mathcal{W}_{\text{ERA}}$, $v \in \mathcal{W}_{\text{Unsubmitted}}$, and u cites v.

Institutional attribution concordance is evaluated across 363,100 ERA canonical works possessing resolved author affiliation records. Each work is classified into one of three mutually exclusive categories:

1. **Submitting HEP Printed**: The submitting university matches at least one author affiliation string printed on the published work.
2. **Other Australian Entity**: The submitting university is absent, but another Australian entity (an alternate university, CSIRO, or an independent medical institute) is listed.
3. **Pure Foreign Affiliation**: No Australian institutional entity appears in any author affiliation on the publication.

Temporal attribution trends are measured by evaluating concordance shares annually across $2011 \leq t \leq 2016$.

3 Results

3.1 The Inclusion–Omission Landscape

Between 2011 and 2016, OpenAlex indexes 744,574 distinct scholarly outputs carrying an author affiliation associated with at least one of the 40 Australian universities. Table **??** partitions this address-defined corpus by ERA 2018 submission status.

Table 1 Comparative Profile of Australian University Publications in OpenAlex (2011–2016) by ERA Submission Status.

Status	Total Works	Share (%)	Mean Cites	Median Cites	Has DOI (%)	Paratext (%)
In ERA Corpus	401,261	53.9%	66.4	28.0	96.8%	0.02%
Not in ERA Corpus	343,313	46.1%	19.4	2.0	90.5%	0.69%
Total Australian HEP	744,574	100.0%	44.7	11.0	93.9%	0.33%

Of the 744,574 address-attributed outputs, 401,261 works (53.9%) appear in the ERA submission dataset, while 343,313 works (46.1%) were omitted from submission

by all Australian institutions. Submitted works display a median citation count of 28.0 ($\mu = 66.4$), whereas unsubmitted works exhibit a median citation count of 2.0 ($\mu = 19.4$). Within the unsubmitted partition, 41.3% of works record zero or one citation.

Table **??** stratifies the unsubmitted corpus across OpenAlex document types.

Table 2 Document Type Distribution of Australian University Publications in OpenAlex (2011–2016) by ERA Inclusion.

OpenAlex Work Type	In ERA	Not in ERA	Total Works	In-ERA Share (%)
Standard Evaluative Types				
Article	256,377	169,636	426,013	60.2%
Conference Paper	26,614	20,032	46,646	57.1%
Book Chapter	9,747	29,284	39,031	25.0%
Review	5,551	2,477	8,028	69.1%
Book	990	3,476	4,466	22.2%
Formally Ineligible Types				
Other	1,102	28,668	29,770	3.7%
Preprint	1,453	12,571	14,024	10.4%
Conference Abstract	137	6,620	6,757	2.0%
Book Review	72	5,917	5,989	1.2%
Reference Entry	289	4,630	4,919	5.9%
Dataset	8	4,214	4,222	0.2%
Editorial	179	3,457	3,636	4.9%
Peer Review	5	2,236	2,241	0.2%
Paratext / Erratum	92	3,463	3,555	2.6%
Dissertation / Thesis	6	1,318	1,324	0.5%
Letter / Report / Misc	171	2,164	2,335	7.3%
Total	401,261	343,313	744,574	53.9%

Formally ineligible document types (preprints, abstracts, book reviews, editorials, paratext, datasets, and unclassified items) total 75,348 unsubmitted works. Conversely, standard peer-reviewed formats constitute 222,428 unsubmitted works: 169,636 journal articles, 29,284 book chapters, 20,032 conference papers, and 3,476 books. For book chapters and complete books, the proportion submitted to ERA drops to 25.0% and 22.2% respectively.

3.2 Citation Integration of Omitted Australian Works

Cross-referencing the 343,313 unsubmitted publications against the citation graph of ERA-submitted works identifies 45,822 unsubmitted works directly cited by at least one ERA submission. Table **??** disaggregates these internal citation flows.

ERA submissions generate 109,443 directed citation links into unsubmitted Australian publications. Within this subset, the 35,023 journal articles record a global mean citation count of 63.1 (median 25.0), and the 991 books average 104.5 global citations (median 42.0).

Table 3 Unsubmitted Australian University Publications (2011–2016) Directly Cited by ERA-Submitted Papers.

Work Type	Cited Works (N)	Citations from ERA	Global Mean Cites	Global Median Cites
Article	35,023	82,914	63.1	25.0
Conference Paper	3,225	7,108	17.7	8.0
Book Chapter	2,132	4,812	28.7	10.0
Preprint	1,562	3,119	21.5	7.0
Book	991	3,420	104.5	42.0
Review Article	628	2,845	150.0	72.0
Editorial / Other	2,261	5,225	32.1	10.0
Total	45,822	109,443	61.4	22.0

Table **??** documents the five most-cited unsubmitted publications bearing Australian university affiliations.

Table 4 Exemplary Highly Cited Australian Publications (2011–2016) Omitted from ERA Submissions.

DOI	Year	Type	Citations	Publication Title / Primary Affiliation
10.1093/nar/gkv007	2015	Article	40,101	*limma powers differential expression analyses...* (U Melb
10.2753/mtp1069-6679190202	2011	Article	21,450	*PLS-SEM: Indeed a Silver Bullet* (U Newcastle)
10.1016/j.advengsoft.2013.12.007	2014	Article	19,343	*Grey Wolf Optimizer* (Griffith U)
10.1016/j.advengsoft.2016.01.008	2016	Article	14,581	*The Whale Optimization Algorithm* (Griffith U)
10.7717/peerj.453	2014	Article	6,735	*scikit-image: image processing in Python* (U Melbourne)

3.3 Institutional Attribution Discordance and the Temporal Mobility Gradient

Evaluating institutional attribution across 363,100 ERA canonical works possessing author affiliation strings reveals systemic divergence between the submitting institution and the published affiliation. Table **??** details this distribution.

Table 5 Concordance Between ERA Submitting Institution and Published Affiliations Across 363,100 Linked Works.

Affiliation Classification on Published Work	Total Works	Share (%)
Submitting HEP Printed on Work	245,601	67.64%
Submitting HEP Absent, but Other Australian Entity Present	67,537	18.60%
No Australian Entity on Work (Pure Foreign Affiliation)	49,962	13.76%
Total Evaluated Works	363,100	100.00%

The submitting university appears in the author affiliation block in 245,601 works (67.64%). In 67,537 works (18.60%), the submitting HEP is absent while another Australian institutional entity is present. In 49,962 works (13.76%), the publication contains zero Australian author affiliations, listing foreign institutions exclusively.

Table **??** disaggregates institutional attribution concordance by publication year across the six-year retrospective assessment window.

Table 6 Temporal Mobility Gradient: Concordance Between Submitting HEP and Published Affiliation by Publication Year.

Publication Year	Works Evaluated	Submitting HEP (%)	Other AU Entity (%)	Pure Foreign (%)
2016	65,736	74.58%	17.02%	8.40%
2015	65,503	71.69%	17.66%	10.66%
2014	64,328	68.56%	18.40%	13.05%
2013	60,534	65.40%	19.28%	15.32%
2012	55,586	62.64%	19.72%	17.64%
2011	51,406	60.49%	20.07%	19.44%
All (2011–2016)	363,100	67.64%	18.60%	13.76%

Attribution concordance displays a strict monotonic temporal trajectory. For publications in 2016 (one year prior to census), 74.58% carry the submitting HEP affiliation and 8.40% carry purely foreign affiliations. For publications in 2011 (six years prior to census), the submitting HEP share drops to 60.49% while the pure foreign share increases to 19.44%, representing a 2.31-fold expansion in non-domestic attribution.

4 Discussion

4.1 Corpus Definition Boundaries in Olensky's Data Accuracy Framework

Olensky established that bibliometric data accuracy is governed by three hierarchical tiers: transcription errors within raw source records, algorithmic failure during entity disambiguation and citation matching, and structural divergence in corpus definition boundaries [**?** **?**]. While algorithmic matching in modern open infrastructures achieves high recall when identifiers are present [**?**], our empirical findings demonstrate that the dominant source of bibliometric disparity is structural corpus boundary divergence.

The divergence between the ERA 2018 submission dataset (401,261 works) and the OpenAlex Australian address-defined corpus (744,574 works) does not represent database indexing omissions or bibliographic matching error. Rather, it reflects the mathematical confrontation between two fundamentally distinct ontological entities: an address-based production model and a census-based workforce evaluation model.

4.2 Workforce Portability and the Temporal Mobility Gradient

National evaluation frameworks assess the retrospective productivity of eligible academic staff employed at a fixed administrative census date (31 March 2017 for ERA 2018). Under ARC portability provisions, researchers who change institutions prior to the census date submit their complete six-year retrospective publication portfolio under their new employing institution, regardless of where the research was performed or funded [**?**].

This administrative mechanism generates the monotonic temporal gradient observed in Table **??**. The probability that an author was employed at their census-date institution decays monotonically with temporal distance from the census date. Consequently, works published in 2011 exhibit a 19.44% pure foreign affiliation rate and a 20.07% inter-institutional domestic divergence rate, meaning that 39.51% of submitted 2011 outputs carry no published affiliation with the university claiming credit for the work. University rankings that aggregate citations from address-indexed databases measure geographical site production, whereas national evaluation exercises measure the mobile intellectual capital accumulated at a single census date [**?**].

4.3 Trainee Authorship and Pre-Census Attrition

Doctoral and postdoctoral researchers author substantial volumes of high-impact scholarship within university laboratories. ERA eligibility regulations restrict submission rights to staff holding formal continuing or fixed-term academic appointments of at least 0.4 FTE at the census date [**?**].

When graduate researchers complete their candidatures and transition to industry, clinical practice, or overseas postdocs prior to the census date, their publications become administratively unclaimable by the host institution. This mechanism accounts for the omission of foundational computational algorithms developed by enrolled doctoral students (such as *Grey Wolf Optimizer*, cited over 19,000 times), despite explicit university affiliations on the published records.

4.4 Independent Research Institutes and External Payroll Boundaries

In biomedical and physical sciences, extensive research activity is situated within independent Medical Research Institutes (MRIs) and statutory government agencies (e.g., CSIRO, ANSTO). Authors affiliated with these organizations routinely maintain adjunct or honorary university titles.

ARC guidelines prohibit universities from claiming outputs authored by honorary staff unless the university paid a substantive academic salary or established audited joint-appointment cost-sharing contracts [**?**]. Consequently, landmark biomedical software contributions (exemplified by *limma*, with over 40,000 citations) are excluded from university submissions despite explicitly displaying university departmental addresses on the publication masthead.

4.5 Internal Citation Integration and Copied Reference Dynamics

The unsubmitted corpus is not scientifically marginal. ERA-submitted publications generate 109,443 direct citation links into 45,822 unsubmitted Australian works. Within academic citation networks, highly cited papers propagate through secondary citation copying without independent verification, magnifying visibility disparities between submitted and omitted literature [**?** **?**]. When performance evaluations restrict citation baselines to submitted sets, they truncate the empirical reference graph, systematically distorting disciplinary field-normalized benchmarks [**?** **?**].

5 Conclusion

Evaluative bibliometrics and global university ranking systems frequently conflate national research assessment corpora with address-indexed bibliographic databases. Linking the comprehensive submission dataset of Australia's ERA 2018 assessment against OpenAlex refutes this equivalence. Across 2011–2016, 46.1% (343,313 works) of all publications bearing an Australian university author affiliation in OpenAlex were absent from the national assessment submission. This omitted corpus includes 222,428 standard peer-reviewed outputs: 169,636 journal articles, 29,284 book chapters, 20,032 conference papers, and 3,476 books.

These omitted works are intrinsically integrated into the national scientific literature: ERA submissions direct 109,443 citations into 45,822 unsubmitted Australian works, encompassing foundational algorithmic and methodological contributions recording tens of thousands of global citations. Furthermore, among submitted works, the institution claiming the paper in ERA matches the published institutional address on the work in only 67.64% of cases. A total of 49,962 submitted works (13.76%) carry zero Australian institutional affiliations on the published document. This discordance follows a strict monotonic temporal gradient across the six-year evaluation window, with pure foreign attribution expanding from 8.40% in 2016 to 19.44% in 2011.

These disparities are not artifacts of bibliographic indexing failure or imperfect string matching. Grounded in Olensky's taxonomy of data accuracy in bibliometric sources [**?**], these patterns demonstrate the structural divergence between two incompatible bibliometric constructs: an **address-based production model** measuring research performed at physical institutional sites, and a **census-based workforce evaluation model** assessing the mobile intellectual output of personnel employed on a single statutory day. Evaluative exercises and citation impact studies must explicitly account for workforce portability, student attrition, and hospital/institute payroll boundaries rather than treating assessment submissions as complete proxies for institutional research output.

Acknowledgments. The author acknowledges access to the Australian Research Council ERA 2018 public repository data and the OpenAlex open scholarly data infrastructure.

Declarations

The author declares that they have no competing interests.

References

[] Hicks, D.: Performance-based university research funding systems. Research Policy **41**(2), 251–261 (2012)

[] Van Raan, A.F.: The use of bibliometric analy-sis in research performance assessment and monitoring of interdisciplinary scientific de-velopments (2003)

[] Van Raan, A.F.J.: Fatal attraction: Conceptual and methodological problems in the ranking of universities by bibliometric methods. Scientometrics **62**(1), 133–143 (2005) https://doi.org/10.1007/s11192-005-0008-6

[] Olensky, M.: Data accuracy in bibliometric data sources and its impact on citation matching. PhD thesis, Humboldt-Universität zu Berlin, Berlin, Germany (2015). https://doi.org/10.18452/17122

[] Olensky, M., Schmidt, M., van Eck, N.J.: Evaluation of the citation matching algorithms of cwts and ifq in comparison to the web of science. Journal of the Association for Information Science and Technology **67**(10), 2550–2564 (2016) https://doi.org/10.1002/asi.23590

[] Mow, K.: Inside the Black Box: Research Grant Funding and Peer Review in Australian Research Councils. Lambert Academic Publishing, ??? (2010)

[] Priem, J., Piwowar, H., Orr, R.: OpenAlex: A fully-open index of scholarly works, authors, venues, institutions, and concepts. arXiv (2022). https://doi.org/10.48550/arXiv.2205.01833

[] Simkin, M.V., Roychowdhury, V.P.: Copied citations create renowned papers? arXiv preprint cond-mat/0305150 (2003) arXiv:cond-mat/0305150

[] Simkin, M.V., Roychowdhury, V.P.: Read before you cite! arXiv preprint cond-mat/0212043 (2002) arXiv:cond-mat/0212043

[] Hicks, D., Wouters, P., Waltman, L., De Rijcke, S., Rafols, I.: Bibliometrics: The leiden manifesto for research metrics. Nature **520**(7548), 429–431 (2015) https://doi.org/10.1038/520429a

[] Mitchell, M., Johnson, S.L., Porter, M.D.: Affiliation errors and distorted researcher mobility: evidence from openalex and scopus. Scientometrics **131**(7), 4743–4762 (2026) https://doi.org/10.1007/s11192-026-05689-9